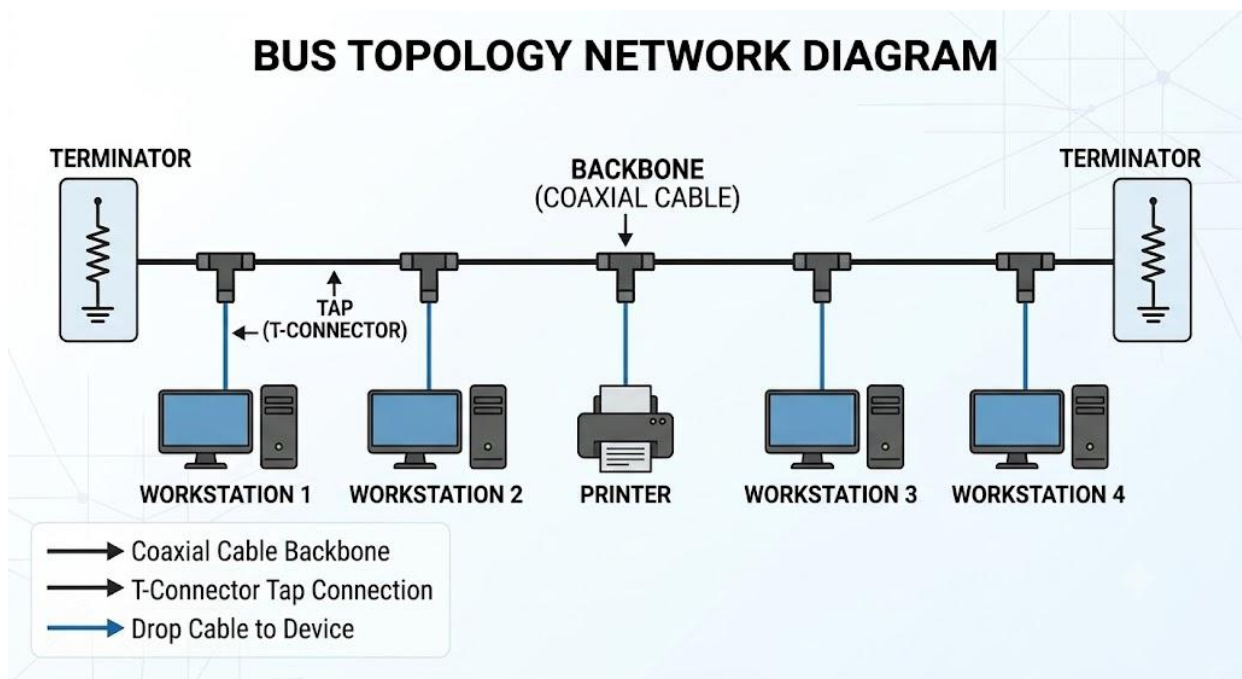


# ASSIGNMENT 2.1

## Network Topologies Theory

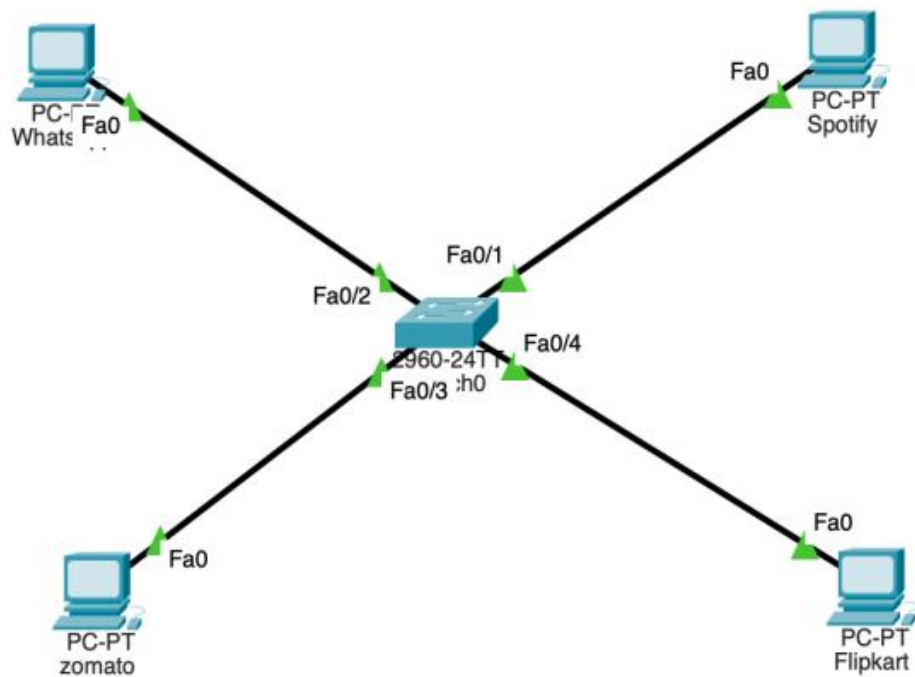
### Task (1):



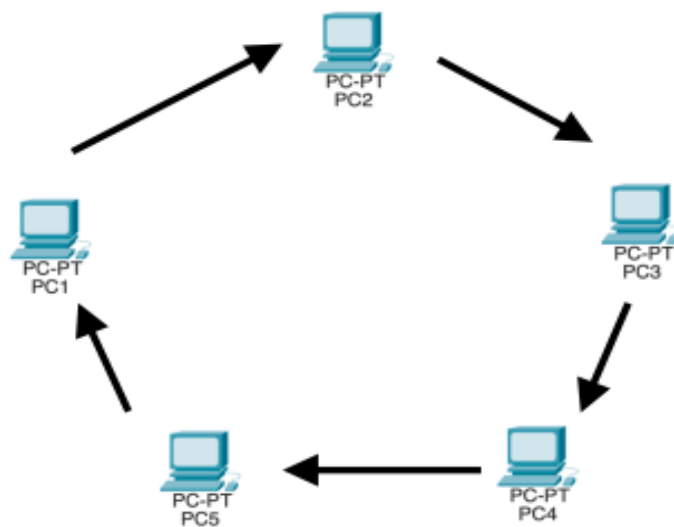
### Real-World Example:

Bus topology was commonly used in older computer networks in schools or small offices, where several computers were connected to one main coaxial cable.

## Task (2):



## Task (3):



## Task (4):

### Mesh vs Star Topology:

| Topology             | Advantages                                                                                                                        | Disadvantages                                                                                                            |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Mesh Topology</b> | 1. Very <b>reliable</b> because devices have multiple connections.2. If one connection fails, the network can <b>still work</b> . | 1. Needs <b>more cables</b> , so it is expensive.2. <b>Difficult to install and maintain</b> .                           |
| <b>Star Topology</b> | 1. <b>Easy to install and manage</b> .2. If one device/cable fails, the <b>other devices keep working</b> .                       | 1. If the <b>main hub/switch fails</b> , the whole network stops.2. Needs <b>more cable</b> than some simple topologies. |